

# Lab # 12

## Introduction to Test Suite

### Lab Task:-

- Make a class having four functions for determining,
  - Whether the input integer is odd.
  - Whether the input integer is even.
  - Whether the input integer is prime.
  - For calculating the factorial of that input integer.

Write their test cases and execute them in a single test suite class. Follow all the steps mentioned above in the manual.

### Code:-

```
1 package Lab12;
2 public class NumberUtils {
3     public boolean isOdd(int n) {
4         return n % 2 != 0;
5     }
6     public boolean isEven(int n) {
7         return n % 2 == 0;
8     }
9     public boolean isPrime(int n) {
10        if (n <= 1)
11            return false;
12
13        for (int i = 2; i <= Math.sqrt(n); i++) {
14            if (n % i == 0)
15                return false;
16        }
17        return true;
18    }
19    public long factorial(int n) {
20        long fact = 1;
21        for (int i = 1; i <= n; i++) {
22            fact = fact * i;
23        }
24        return fact;
25    }
26 }
```

```
1 package Lab12;
2 import org.junit.Test;
3 import static org.junit.Assert.*;
4 import org.junit.Test;
5 public class NumberUtilsTest {
6     NumberUtils obj = new NumberUtils();
7     @Test
8     public void testOdd() {
9         assertTrue(obj.isOdd(5));
10        assertFalse(obj.isOdd(10));
11    }
12    @Test
13    public void testEven() {
14        assertTrue(obj.isEven(10));
15        assertFalse(obj.isEven(7));
16    }
17    @Test
18    public void testPrime() {
19        assertTrue(obj.isPrime(7));
20        assertFalse(obj.isPrime(9));
21    }
22    @Test
23    public void testFactorial() {
24        assertEquals(120, obj.factorial(5));
25        assertEquals(1, obj.factorial(0));
26    }
27 }

1 package Lab12;
2 import org.junit.platform.suite.api.SelectClasses;
3 import org.junit.platform.suite.api.Suite;
4 @Suite
5 @SelectClasses({ NumberUtilsTest.class })
6 public class AllTests {
7
8 }
9
```

Output:-



